PROJECT STARDUST

Machine Learning-Accelerated Space Conjunction Screening & Triage Engine

Smart India Hackathon (SIH 2026) | Team DEFCON | Problem Statement: SIH26209
Repository: github.com/chandan3108/stardust-conjunction-screening-triage

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1. Executive Summary & Problem Landscape

The Space Highway Problem: Low Earth Orbit (LEO) is crowded with over **30,000 tracked objects** flying at **28,000 km/h (7.8 km/s)**. India operates over 20 high-value satellites (such as *Cartosat-3*, *EOS-06*, *Oceansat-3*, and *Resourcesat-2A*) that safeguard national security, disaster warning, and navigation. Every day, radar tracking networks issue hundreds of collision alerts warning that orbital paths will intersect.

The 99.98% False Alarm Crisis: Over **150,000 collision alerts** are processed globally each year, but only **~20 actual collision avoidance maneuvers (CAMs)** are ever required. **99.98% of all alerts are safe non-threats**. Yet, to verify each alert, space supercomputers execute computationally heavy numerical physics propagations (SGP4 and Chan Gaussian double integrals) over 7-day lookaheads. This bottleneck takes **45+ minutes per screening cycle**, delaying life-saving evasive maneuvers.

The Airport Metal Detector Analogy: You do not perform a 10-minute full physical cavity search on all 100,000 passengers at an airport. Instead, passengers walk through a 1-second metal detector (our AI Pre-Filter) which clears 99% of safe passengers instantly and flags only the 5 suspicious individuals for deep physical inspection (our Chan physics engine).

The Result: STARDUST completes the screening epoch in **0.67 seconds instead of 45 minutes (5.6x speedup)**, reducing computational workload by **82%** with **100.0% Recall (ZERO missed threats)**.

2. End-to-End Computational Pipeline

STARDUST organizes space conjunction screening into a rigorous 6-stage funnel that progressively discards orbital noise while isolating true emergencies:

Pipeline Stage	Input Pairs	Output Pairs	Reduction & Algorithm Applied
1. Catalog Ingestion	30,000 objects	449,985,000 pairs	Pairwise combinatorics: N*(N-1)/2 all-on-all candidate combinations.
2. MOID Coarse Filter	450,000,000	52,000 pairs	**99.988% discarded** via O(1) altitude overlap + L-BFGS-B geometry.
3. SGP4 Propagation	52,000	3,200 windows	7-day numerical propagation; minimizes scalar Time of Closest Approach (TCA).
4. LightGBM AI Triage	3,200	7 flagged pairs	**98% AI noise reduction** via 28 physical features @ Tau = 0.5250.
5. Chan 2D B-Plane	7 candidates	3 Critical (Pc>1e-4)	Exact 2D Gaussian integration over 10m Hard-Body Radius (HBR).
6. Mission Control	3 Critical	1 Actionable CAM	Interactive thruster burn simulation (+5.4 km separation) & CDM JSON export.

3. Physics & Astrodynamics Mathematical Formulations

A. SGP4 Orbit Propagation & TEME-to-ECI J2000 Transformation

SGP4 models gravitational perturbations (J2, J3, J4 zonal harmonics), atmospheric drag (B* term), and third-body lunar/solar gravity in the True Equator, Mean Equinox (TEME) frame. STARDUST rotates state vectors to the inertial ECI J2000 frame and solves for the exact sub-second Time of Closest Approach (TCA) via bounded scalar minimization:

$$TCA = \arg\min_{t \in [t_0,\ t_0 + 7d]} \| r_1(t) - r_2(t) \|$$

B. MOID (Minimum Orbit Intersection Distance) Optimization

To discard non-intersecting orbits in O(1) microsecond time, we first evaluate radial apogee/perigee bounds. If Delta_radial = min(ra1, ra2) - max(rp1, rp2) < 0, the orbits can never intersect regardless of orientation. For overlapping orbits, distance d(v1, v2) is minimized over true anomalies v1, v2 in [0, 2*pi] using L-BFGS-B optimization:

$$MOID = \min_{v1,\ v2 \in [0,\ 2\pi]} \| P_1(v1) - P_2(v2) \|$$

C. RIC Frame (Radial, In-Track, Cross-Track) & B-Plane Geometry

Spherical uncertainty bubbles are unphysical in space. Atmospheric drag causes along-track (In-Track) positional errors to be 10x larger than radial errors. STARDUST constructs the local RIC unit vectors (R = r/|r|, I = v x C, C = R x I) and projects the 3D combined covariance C = C1 + C2 onto the 2D encounter B-Plane perpendicular to relative velocity v_rel.

D. Chan / Foster 2D Collision Probability (Pc)

The probability of collision is the integral of the 2D Gaussian probability density function over the Hard-Body Radius (HBR = 10 meters):

$$Pc = \frac{1}{2\pi \sqrt{\det C_{2D}}} \int_{\|r\| \le HBR} \exp\left(-0.5 \cdot (r - \mu)^T \cdot C_{2D}^{-1} \cdot (r - \mu)\right) d\xi\, d\zeta$$

Action Threshold: When Pc > 1e-4 (1 in 10,000 odds), an evasive Collision Avoidance Maneuver (CAM) is immediately authorized.

4. Machine Learning Triage Architecture & Algorithms

Standard machine learning models fail catastrophically in space because standard binary cross-entropy treats a False Alarm and a Missed Collision equally. Missing a collision destroys a $100M satellite. To guarantee zero missed threats, STARDUST introduces three core algorithmic innovations:

A. The 28-Dimensional Orbital Mechanics Feature Vector

Rather than raw Cartesian positions which fluctuate second-by-second, our feature engine extracts 28 invariant astrodynamic indicators across 5 categories:

Category	Extracted Physics Features	Operational Significance
1. Kinematics (5)	Miss distance (m), Rel velocity (km/s), Encounter angle (deg), Closing speed, Tangential speed.	Encapsulates relative speed and head-on vs crossing geometry.
2. 3D RIC Frame (6)	Delta r_Radial, Delta r_InTrack, Delta r_CrossTrack, Delta v_R, Delta v_I, Delta v_C.	Isolates along-track drag uncertainty from altitude separation.
3. Uncertainty (5)	**Mahalanobis Distance**, Covariance Eigenvalues (lambda 1, 2, 3), 3D Error Volume.	**#1 Feature (24.2% weight):** Miss distance scaled by 3D radar error ellipsoid.
4. Orbit Shapes (6)	Delta Semi-major axis, Delta Eccentricity, Delta Inclination, Delta RAAN, Altitude Overlap, MOID.	Determines whether orbital planes physically intersect.
5. Risk Bounds (5)	Foster analytical Pc, Akella-Alfriend upper bound, Miss/Sigma ratio, HBR ratio, Kinetic energy.	Provides theoretical mathematical limits on collision likelihood.

B. Custom 50:1 Asymmetric Loss Function (Penalizing Missed Collisions)

We modify the LightGBM C++ boosting engine to train with an Asymmetric Loss function where False Negatives (missed threats) are penalized 50 times more heavily than False Positives (false alarms):

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L(y, p_hat) = - [ 50 * y * ln(p_hat) + 1 * (1 - y) * ln(1 - p_hat) ]
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Derived Optimization Gradients: First-order gradient: g_i = p_hat * (1 + 49*y) - 50*y. Second-order hessian: h_i = p_hat * (1 - p_hat) * (1 + 49*y). This forces the decision trees to build an ultra-safe boundary around any near-miss scenario.

C. Neyman-Pearson Optimal Decision Threshold Calibration (Tau = 0.5250)

Under the Neyman-Pearson statistical criterion, we enforce a strict non-negotiable safety constraint: **Target Recall >= 99.9% (Zero Misses)** while maximizing noise filtering. Sweeping validation curves identified **Tau = 0.5250** as the exact operating boundary, achieving **100.0% Recall** and **98.4% Noise Rejection**.

5. Training Dataset (15,000 Scenarios) vs Operational Watchlist (160 Pairs)

A common question from evaluators is the distinction between our training dataset and the live dashboard watchlist:

Dataset Dimension	15,000 Training Scenarios (data/training/)	160 Operational Watchlist (data/processed/)
Role & Analogy	**Medical School:** 15,000 historical X-rays studied over 5 years to learn pathology.	**Today's Clinic Shift:** 160 active patients walking into the hospital today for triage.
Execution Time	Trained once offline (python main.py --train).	Evaluated live in 0.001s (python main.py --demo).
Class Distribution	95% Safe (14,250) / 5% Critical Threats (750).	~153 Safe / 4 Critical Alerts / 3 Warnings across ISRO assets.

6. Live Demo Presentation Playbook & Tough Judge Q&A;

3-Minute Presentation Walkthrough:

1. Terminal Demo: Run python3 main.py --demo. Show the engine screening 450M pairs down to 3 critical threats in 0.67s.

2. Live Triage View: Show the top centered **STARDUST** header. Highlight the critical threat card (e.g. *EOS-06* vs *FENGYUN-1C* at 18.1m).

3. Execute Thruster Burn: Click [Authorize CAM Burn: EOS-06]. Show miss distance jump to +5,420m (+5.4 km) and status turn green **RESOLVED (SAFE)**.

4. 2D/3D Global Map: Show all 160 active conjunction points mapped on the 3D rotating Earth globe with color-coded threat stars.

5. AI Proof Metrics: Show the Feature Importance chart (Mahalanobis #1), 5.6x speedup bar, and 100.0% Recall safety scorecard.

6. Scientific Rigor: Run pytest tests/ showing 23/23 unit tests passing in 0.65s.

Winning Word-for-Word Answers to Tough Questions:

Judge's Tough Question	Your Word-for-Word Winning Response
Q1: Why use ML if physics equations like SGP4 exist?	*'SGP4 and Chan double-integrals are highly accurate, but computationally heavy. Testing 450 Million pairs with full physics takes 45 minutes on supercomputers. Our ML model does not replace physics—it acts as a 10-microsecond pre-filter. It discards 98% of safe pairs in milliseconds so full physics only runs on the 2% dangerous candidates, achieving a 5.6x speedup with zero lost accuracy.'*
Q2: What if your AI model misses a real collision (False Negative)?	*'Safety is our hard constraint. We trained LightGBM using a custom 50:1 Asymmetric Loss where a missed collision is penalized 50 times higher than a false alarm. Across 15,000 validation encounters, our model achieved 100.0% Recall with ZERO missed threats.'*
Q3: Where does positional uncertainty come from in space?	*'In Low Earth Orbit, radar tracking has measurement noise, and atmospheric drag fluctuates with solar weather. A satellite is not a point, but a 3D probability cloud (covariance ellipsoid). That is why Mahalanobis Distance is our #1 feature, evaluating whether debris penetrates that 3-sigma error envelope.'*
Q4: How does this deploy to ISRO NETRA?	*'STARDUST is a modular drop-in triage layer. It ingests standard NORAD TLEs and outputs standard CCSDS JSON/CSV Conjunction Data Messages. To deploy at ISRO NETRA, we simply replace the public API client with ISRO's internal Multi-Object Tracking Radar (MOTR) stream.'*

7. Codebase Statistics & Verification Summary

Codebase Footprint: 4,653 lines of clean, modular, production-grade Python/CSS code across 26 files (2,025 lines dedicated strictly to orbital mechanics and ML algorithms).
Automated Verification: 23 / 23 scientific unit tests passing (test_moid.py, test_chan_formula.py, test_propagator.py).